

Nicholas Donovan: <https://www.linkedin.com/in/nicholas-d-infosec23/>

Analyzing Redline Pcap

Summary:

According to Sophos, Squirrelwaffle is a malware loader that is distributed as a malicious Office document in spam campaigns. It provides attackers with an initial foothold in a victim's environment and a channel to deliver and infect systems with other malware. When a recipient opens a Squirrelwaffle-infected document and enables macros, a visual basic script typically downloads and executes malicious files and scripts, giving further control of the computer to an attacker. Squirrelwaffle operators also use DocuSign to try and trick the user into enabling macros in Office documents.

Debugging

Clicking the malware sample link will bring you to this page. You can make an account and

[Download](#)

Hashes

MD5: dd6257665f634b5566e15bc62e90c809

SHA1: 54ded03f7b82c8aa4de10e673b556d0ba778177d

SHA256: 6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab

SSDEEP: 1536:CZYCKkPkDwBycMjIBbzreg/+UgOiALGr05:CZYCKkPkc7bHpdpin05

Observed File Names

squirrel.bin

Yara Hits

YRP/Microsoft_Visual_Cpp_v50v60_MFC	YRP/Borland_Delphi_30_additional	YRP/Borland_Delphi_30_	YRP/Borland_Delphi_v40_v50
YRP/Borland_Delphi_v30	YRP/Borland_Delphi_DLL	YRP/IsPE32	YRP/IsDLL
YRP/IsWindowsGUI	YRP/HasDebugData	YRP/HasRichSignature	
YRP/domain	YRP/contentis_base64	YRP/anti_dbg	YRP/network_dns
YRP/BASE64_table	YRP/Str_Win32_Winsock2_Library		

VT Context

✕ Detections | IoCs | Graph | Attribution
VT Augment by VIRUSTOTAL

58

/ 74

-1

Community Score

58 security vendors flagged this file as malicious

🔗 [trojan.squirrelwaffle/genericcrxqc](#)

6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab

6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab.exe

pedll
spreader

File type	File size	Last analysis
Win32 DLL	73.7 kB	18 days ago

Full report

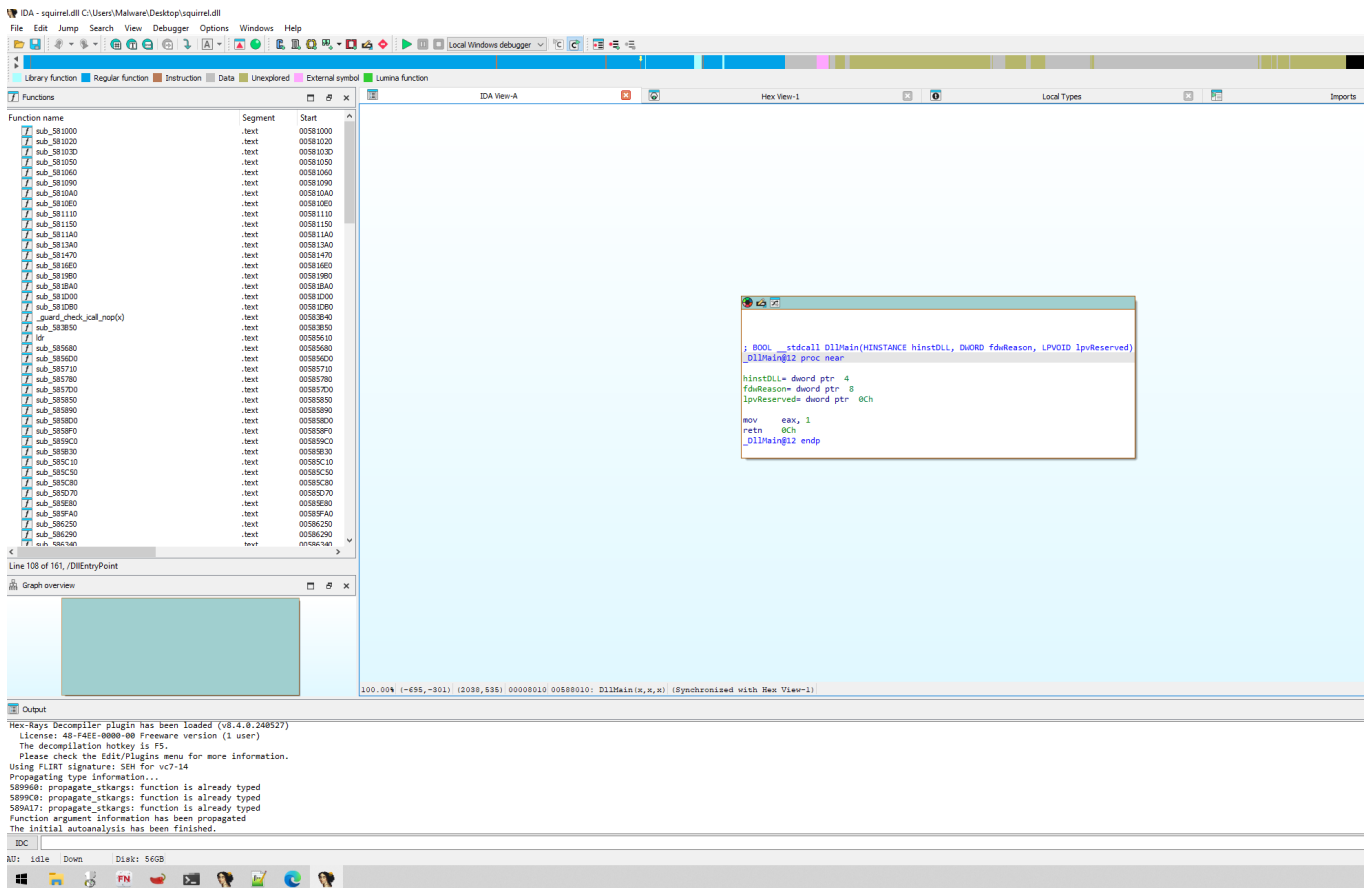
Similar files

VT Graph

SECURITY VENDORS SCANNING RESULTS ❗

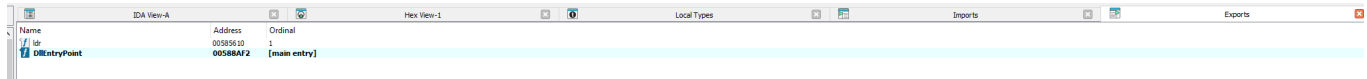
Bkav Pro: W32.AIDetectMalware Elastic: Windows.Trojan.Squirrelwaffle CTX: dll.trojan.generic	Lionic: Trojan.Win32.Squirrelwaffle.alc eScan: Trojan.GenericKD.47072948
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Upon opening IDA click new > select dll file



Click Exports

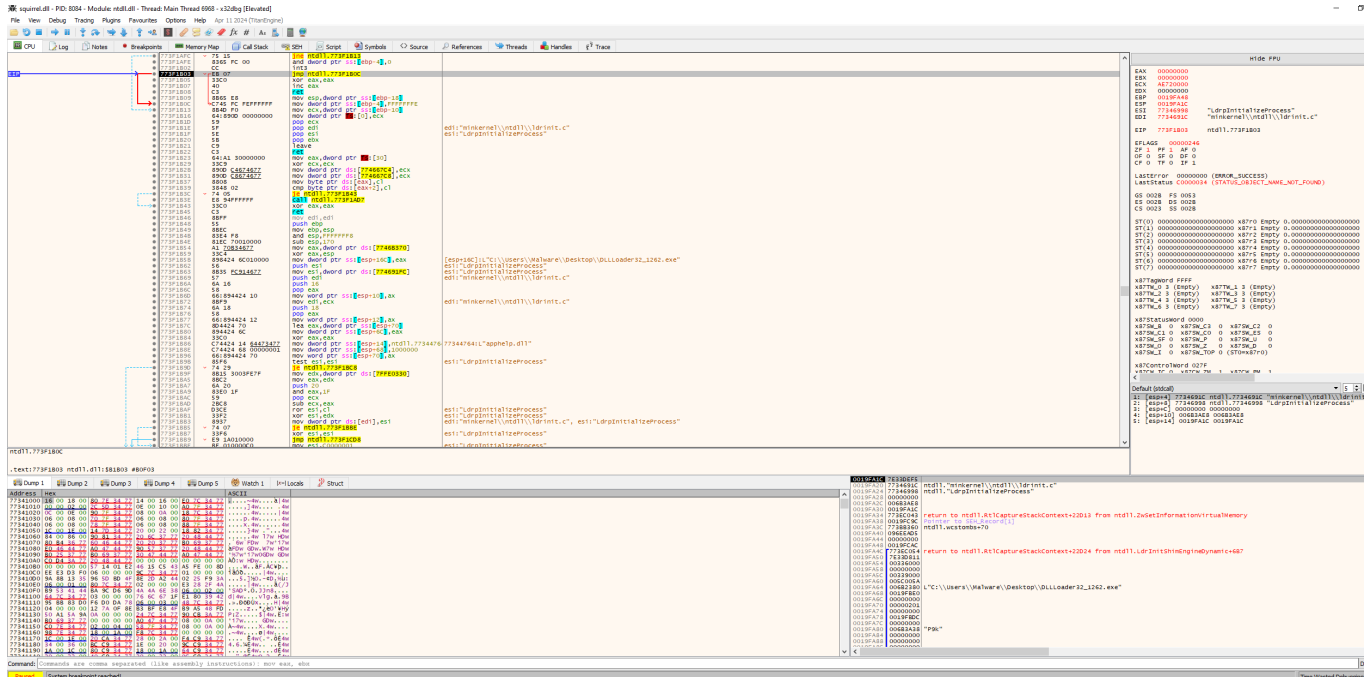
In exports we see two loaders:



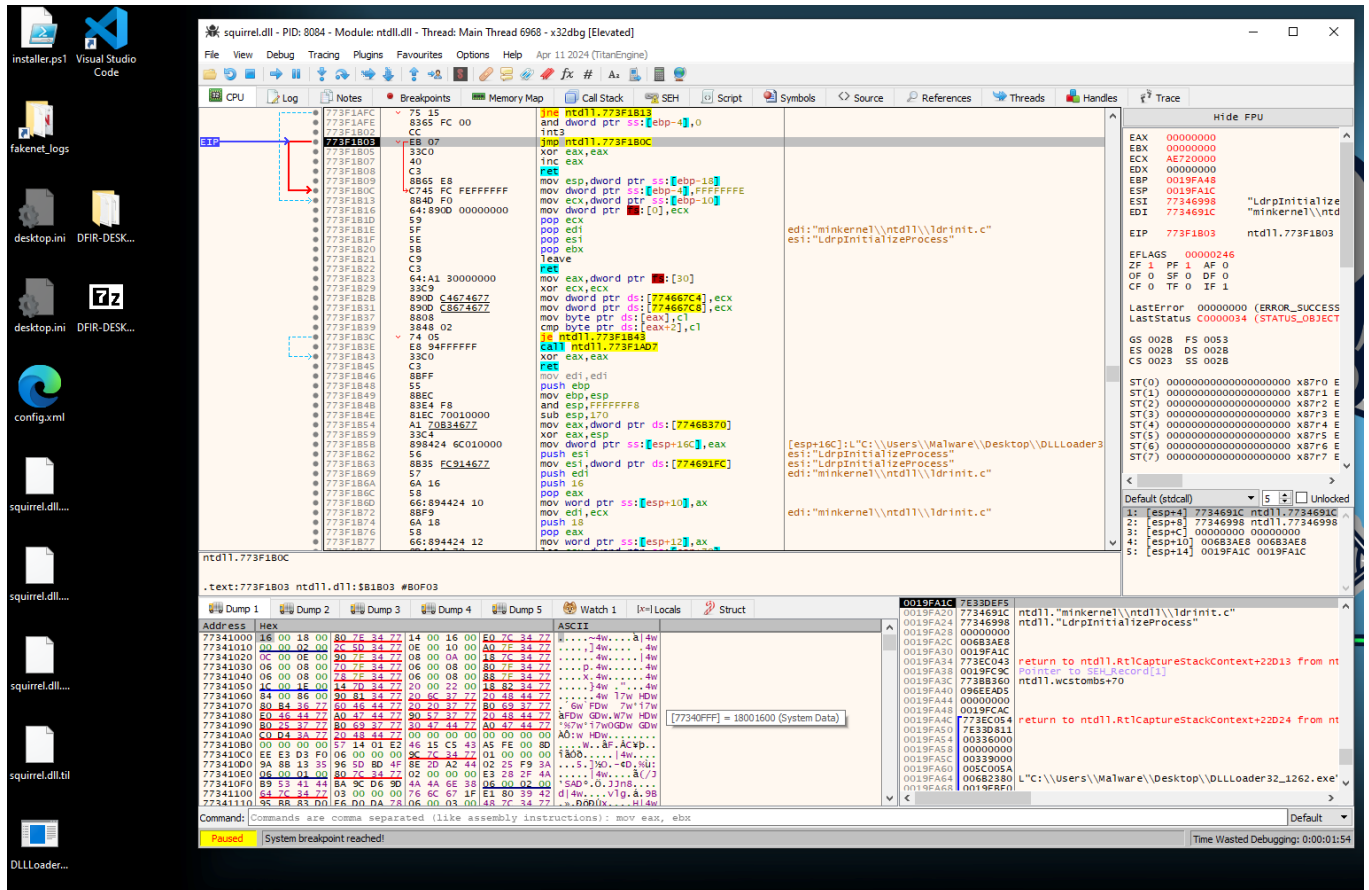
ldr is set as ordinal 1 which is where the config decryption occurs for this dll.

DLLEntryPoint does not have anything interesting stored within it at this time and if we attempted to debug this we wouldn't be able to debug the code within the loader.

Loading the export into x32dbg due to the dll being 32bit.



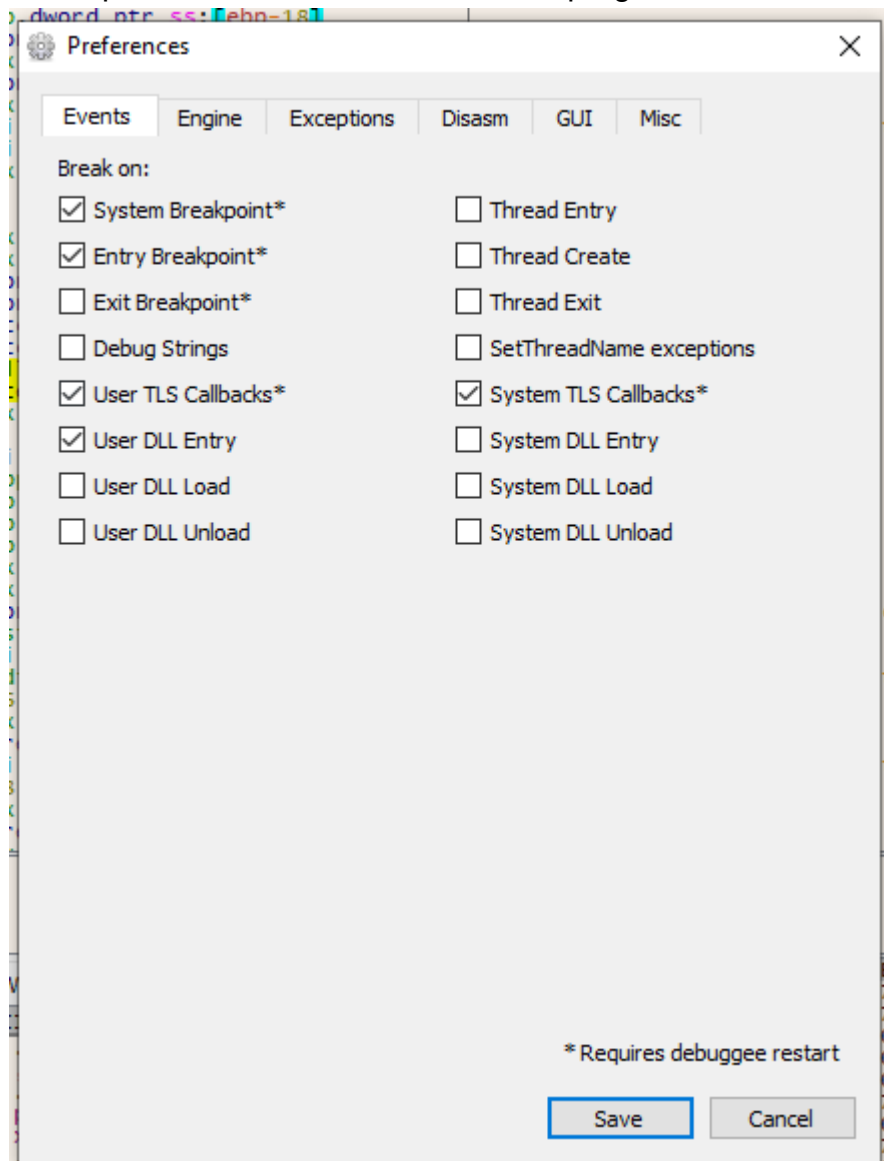
For further testing a PE file is created by x32dbg so that it may load our malicious DLL for analysis.



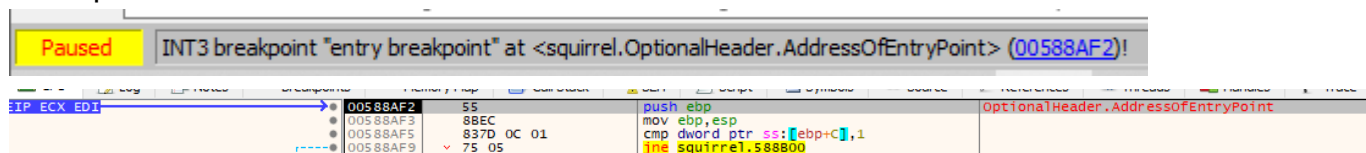
Bottom left ^^

Right now we want to break the entry point of that DLL so that we may access the code behind the loader itself.

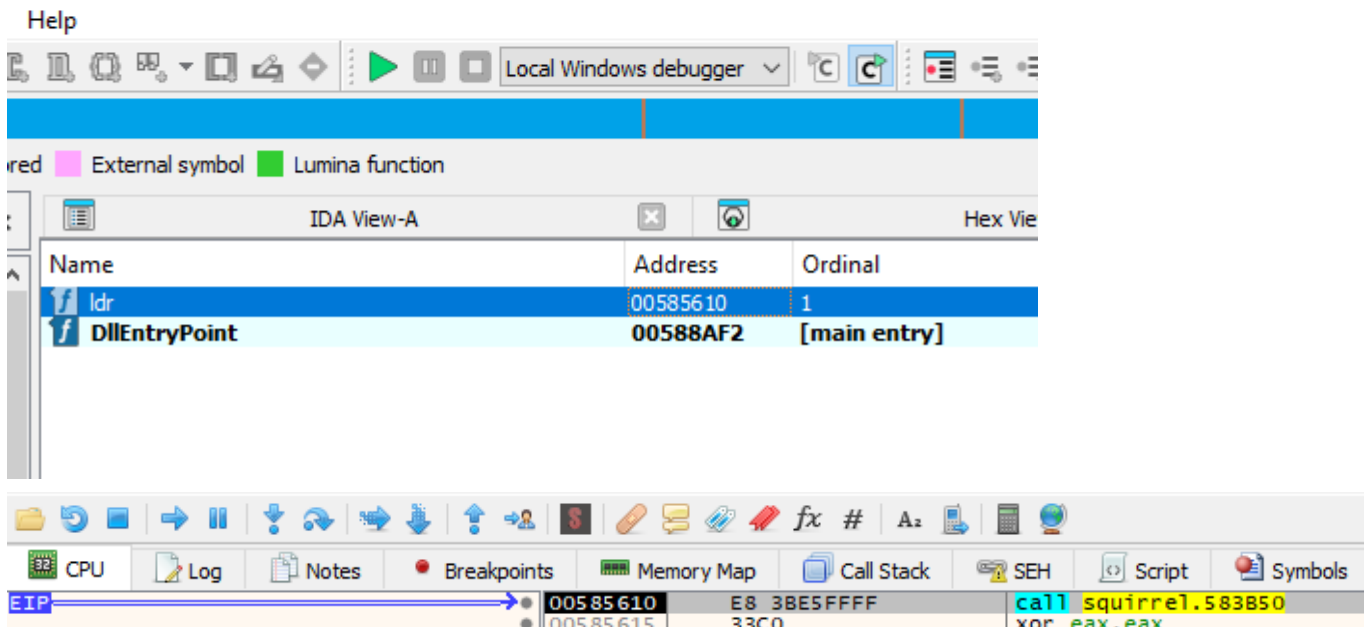
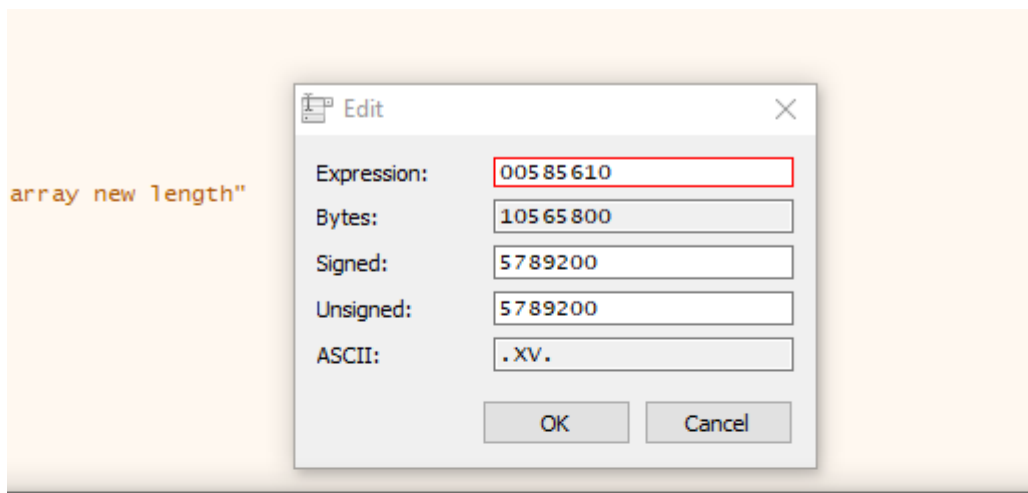
In options you'll want to enable User DLL Entry so that we can set the breakpoint between the two exports. From there we will run the program until we see the breakpoint occur.



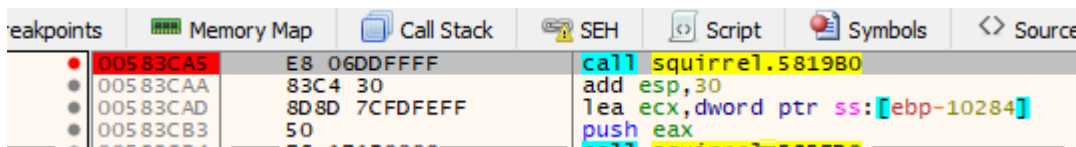
Breakpoint:



Right now the breakpoint is set to the DLLEntryPoint. We need to change the EIP to match the loader (ldr) and that will call the preferred export.



Now we need to set a new breakpoint at the decryption portion here and then take a look at the return value of eax.



After this we run the debugger and we right click at the bottom and follow DWORD in current dump which will reveal the blocklist IP addresses of the DLL config.

Address	Hex	ASCII
00FF28C0	39 34 2E 34 36 2E 31 37 39 2E 38 30 0D 0A 32 30	94.46.179.80..20
00FF28D0	36 2E 31 38 39 2E 32 30 35 2E 32 35 31 0D 0A 38	6.189.205.251..8
00FF28E0	38 2E 32 34 32 2E 36 36 2E 34 35 0D 0A 38 35 2E	8.242.66.45..85.
00FF28F0	37 35 2E 31 31 30 2E 32 31 34 0D 0A 38 37 2E 31	75.110.214..87.1
00FF2900	30 34 2E 33 2E 31 33 36 0D 0A 32 30 37 2E 32 34	04.3.136..207.24
00FF2910	34 2E 39 31 2E 31 37 31 0D 0A 34 39 2E 32 33 30	4.91.171..49.230
00FF2920	2E 38 38 2E 31 36 30 0D 0A 39 31 2E 31 34 39 2E	.88.160..91.149.
00FF2930	32 35 32 2E 37 35 0D 0A 39 31 2E 31 34 39 2E 32	252.75..91.149.2
00FF2940	35 32 2E 38 38 0D 0A 39 32 2E 32 31 31 2E 31 30	52.88..92.211.10
00FF2950	39 2E 31 35 32 0D 0A 31 37 38 2E 30 2E 32 35 30	9.152..178.0.250
00FF2960	2E 31 36 38 0D 0A 38 38 2E 36 39 2E 31 36 2E 32	.168..88.69.16.2
00FF2970	33 30 0D 0A 39 35 2E 32 32 33 2E 37 37 2E 31 36	30..95.223.77.16
00FF2980	30 0D 0A 39 39 2E 32 33 34 2E 36 32 2E 32 33 0D	0..99.234.62.23.

Resources:

<https://malpedia.caad.fkie.fraunhofer.de/details/win.squirrelwaffle>

<https://blog.talosintelligence.com/squirrelwaffle-emerges/>

https://www.trendmicro.com/en_us/research/21/k/Squirrelwaffle-Exploits-ProxyShell-and-ProxyLogon-to-Hijack-Email-Chains.html

<https://any.run/malware-trends/squirrelwaffle/>

<https://www.virustotal.com/gui/file/6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab>

<https://attack.mitre.org/software/S1030/>

<https://github.com/mandiant/flare-vm>

Malware Sample: [https://malshare.com/sample.php?](https://malshare.com/sample.php?action=detail&hash=6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab)

[action=detail&hash=6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab](https://malshare.com/sample.php?action=detail&hash=6095f96dd5eca96a3fb9338eec4ab574921c0febb36f6a6db60aae1aeb9ffcab)